Calibrated and Kernel-Based Detection of QKD Implementation Attacks in a Controlled Simulation

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***Abstract*—Machine-learning detectors for attacks on quantum key distribution (QKD) are usually tested on simulated telemetry. Many report near-perfect accuracy. We show that this can come from the simulator. Without drift in normal operation, a single telemetry feature separates attacks perfectly. We build a seeded decoy-state BB84 simulator with operating drift, finite-size noise and one attack-strength parameter s. On it we compare a Gaussian-process classifier (GP), an amplitude-encoding fidelity-kernel SVM (AE-SVM), an RBF-kernel SVM, random forests and a stacked model. Every model is trained on the same data at every training size. At $s = 0.3$ all supervised models reach 0.80 to 0.84 MCC. Random forests are best or tied best. The GP is the least calibrated model. It also gives the worst uncertainty for deferring alerts. AE-SVM matches RBF-SVM. It generalizes to unseen attack types only because one feature carries the input's magnitude. Removing that feature drops its held-out AUC from 0.95 to 0.55. A one-coordinate norm anchor restores it to 0.94.**

***Index Terms*—Quantum key distribution, anomaly detection, Gaussian processes, quantum kernels, calibration, simulation**

I. Introduction

QKD draws its security from physics [1], [2]. Real systems still leak through their hardware. Known attacks include photon-number splitting (PNS) [3], [4], detector blinding [5], [6], Trojan-horse probing [7], [8] and intercept-resend [9].

Telemetry monitoring is one defense. Machine learning has been applied to it [10] and to related QKD control tasks [11]. Two problems make such results hard to judge. Detectors are tested on simulated telemetry, and a simulator with one fixed operating point makes most attacks easy to spot. Quantum models are also often trained on fewer samples than the baselines.

This paper is a controlled simulation study. Our contributions are as follows.

1) A seeded telemetry simulator with drift, leakage tests and an attack-strength axis s.
2) A comparison of five detectors trained on the same data at every size.
3) An explanation, confirmed by ablation, of when the amplitude kernel generalizes to unseen attacks.
4) Evidence that a recalibrated random forest gives better alert deferral than a GP posterior.

II. Telemetry Simulator

Each sample is one key-generation block of a decoy-state BB84 link over 50 km of fiber. For mean photon number μ, transmittance T and detector efficiency η, a Poisson source gives [12]

$$Q_\mu = Y_0 + \left(1 - e^{-\eta T \mu}\right)(1 + p_{\text{ap}}), \tag{1}$$

$$E_\mu = \left[\tfrac{1}{2} Y_0 + e_d \left(1 - e^{-\eta T \mu}\right)(1 + p_{\text{ap}})\right] / Q_\mu. \tag{2}$$

Here Y_0 is the dark-count probability, e_d the misalignment error and $p_{\text{ap}} = 0.005$ the afterpulse probability.

Normal operation drifts from block to block. Loss varies by 0.3 dB and detector efficiency by 3%. Misalignment, dark counts and background light vary log-normally. The two error-rate estimates carry finite-size noise for blocks of 5,000 bits. Each block reports seven measurements.

The four attacks perturb these measurements following their published descriptions [3], [5], [7], [9]. The perturbation sizes are modeling assumptions. Every deviation is multiplied by a strength s. At $s = 1$ the attack uses its default size. As s falls toward 0 it becomes indistinguishable from normal blocks (Fig. 1).

We release two leakage tests with the simulator. Different seeds must give different datasets. No single feature may act as a label. An earlier version of our own pipeline failed both tests. There, the error rate alone separated attacks perfectly and every model looked near-perfect.

III. Detectors and Protocol

Every detector sees the same 8 features (Fig. 2). These are the 7 measurements standardized on the training data and a robust Mahalanobis distance [13] to normal training blocks.

AE-SVM pads a vector with zeros to 2^q entries and normalizes it. Its fidelity kernel is

$$K_A(\mathbf{x}, \mathbf{y}) = |\langle \psi(\mathbf{x}) | \psi(\mathbf{y}) \rangle|^2 = (\hat{\mathbf{x}}^{\mathsf{T}} \hat{\mathbf{y}})^2. \tag{3}$$

This is the squared cosine similarity. It is cheap to compute classically. A PennyLane [14] simulation of the overlap circuit agrees with it to 7×10^{-15}. The circuit takes 26 ms per kernel entry and the closed form 3×10^{-4} ms. All 8 features fit on 3 qubits.

K_A ignores the input's norm, since $K_A(c\mathbf{x}, \mathbf{y}) = K_A(\mathbf{x}, \mathbf{y})$ for any $c \neq 0$. A norm-anchored variant appends a constant c before encoding. We set c to the median training norm.

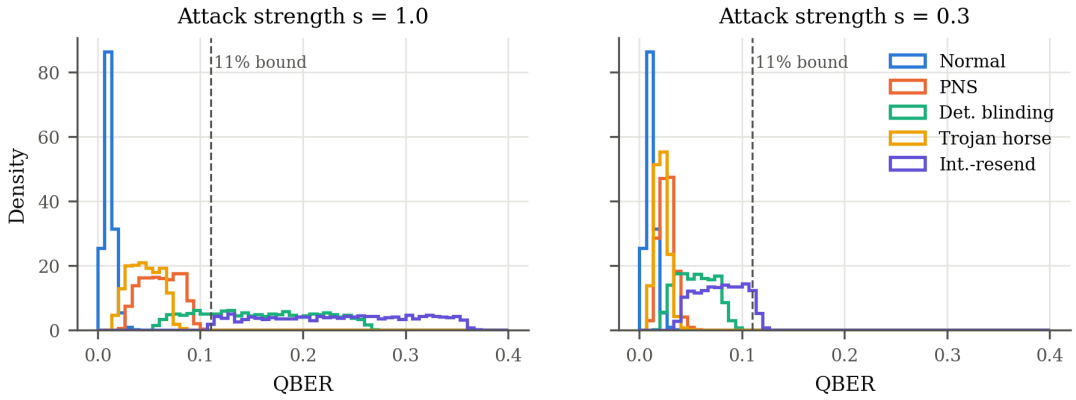


Fig. 1. QBER by class at $s = 1$ and $s = 0.3$. At $s = 0.3$, PNS and Trojan-horse blocks overlap normal operation well below the 11% bound.

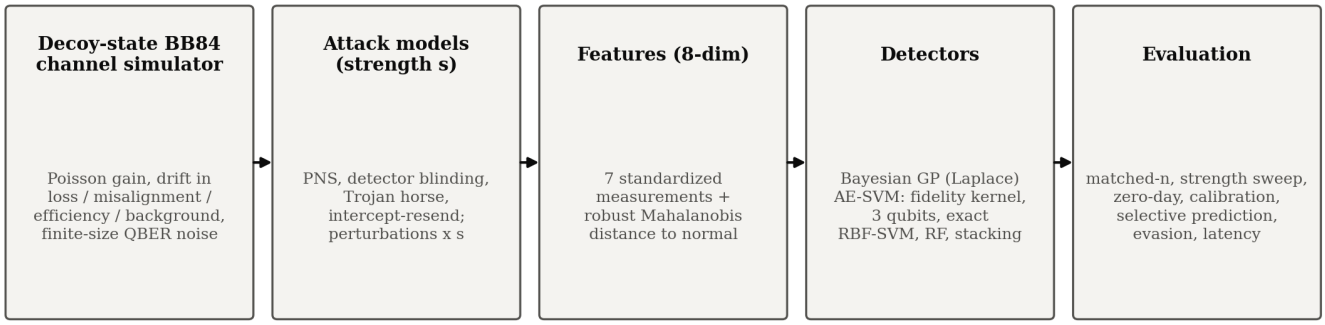


Fig. 2. Study design. All detectors share one 8-dimensional feature map.

The baselines are a Laplace GP [15], an RBF-kernel SVM, random forests (RF, 200 trees) [16] raw and isotonic-calibrated [17], a GP and AE-SVM stacking model (Hybrid) [18] and three novelty detectors fit on normal data only.

Each run uses 2,000 blocks with 30% attacks and a 70/30 split. We repeat it over 10 seeds and report 95% confidence intervals. Matched-size runs train every model on the same subsample. Zero-day runs train on three attack types and test on the fourth.

IV. Results

A. In-distribution detection

At $s = 1$ every supervised model passes 0.95 MCC (Table I). This setting cannot tell the methods apart. At $s = 0.3$ the models spread from 0.80 to 0.84 MCC. RF, isotonic RF and Hybrid lead.

The GP has the lowest attack recall at $s = 0.3$, 0.78 against 0.85 for RF and 0.90 for AE-SVM. It trades recall for specificity. Hybrid fixes 154 GP errors and adds 76 across all seeds. Its gain is significant (McNemar) on 4 of 10 seeds. AE-SVM and RBF-SVM never differ significantly at $s = 0.3$.

B. Training size and attack strength

RF learns fastest at both strengths (Table II). AE-SVM needs more data than RBF-SVM to reach the same MCC. It is never reliably ahead. Hybrid is unstable at $n = 25$, where its stacking folds hold only 8 blocks each.

All four detectors fall together as s shrinks (Table III). RF leads for $s \geq 0.2$. Below that the models are within 0.04 MCC of each other. Blinding and intercept-resend stay visible down to $s = 0.1$ because they raise QBER the most. PNS and Trojan-horse probing are lost first.

C. Unseen attack types

AE-SVM generalizes as well as RF and better than RBF-SVM (Table IV). This is surprising, because its kernel-target alignment [19] is lower. It is 0.13 to 0.22 against 0.29 to 0.66 for RBF.

The explanation is Eq. (3). K_A ignores the norm, but an unseen attack is easiest to spot by how far it sits from normal blocks. In our features that distance is carried by one coordinate, the Mahalanobis distance. Removing it drops AE-SVM to 0.554 at $s = 0.3$ and 0.635 at $s = 1$ (Fig. 3). RBF-SVM is unaffected. The norm anchor restores AE-SVM to 0.939 and 0.980.

TABLE I
DETECTION AT $n = 1400$ (MEAN $\pm$ 95% CI, 10 SEEDS). ECE USES 15 BINS. BOLD MARKS THE BEST SUPERVISED VALUE PER COLUMN.

	$s = 1.0$			$s = 0.3$		
Model	MCC	AUC	ECE	MCC	AUC	ECE
RF	**.975±.008**	**0.999**	0.013	.834±.013	0.969	**0.030**
RF-isotonic	.972±.008	0.997	**0.010**	**.837±.013**	0.967	**0.030**
Hybrid	.971±.005	**0.999**	0.015	.833±.016	0.969	0.037
RBF-SVM	.969±.008	**0.999**	0.011	.810±.015	0.957	0.040
GP	.964±.009	**0.999**	0.039	.801±.016	0.956	0.047
AE-SVM	.955±.011	**0.999**	0.015	.814±.023	**0.970**	0.032
AE-SVM (anchored)	.968±.008	**0.999**	0.012	.816±.019	0.965	**0.030**
OC-SVM	.838±.016	0.994	–	.657±.017	0.839	–
IF	.837±.021	0.997	–	.722±.019	0.929	–
LOF	.824±.021	0.997	–	.691±.011	0.924	–

TABLE II
MCC AT MATCHED TRAINING SIZE n (MEAN OF 10 SEEDS).

s	Model	$n = 25$	100	500	1400
1	AE-SVM	0.795	0.910	0.946	0.955
1	RBF-SVM	0.755	0.920	0.960	0.969
1	GP	0.889	0.916	0.960	0.964
1	RF	0.930	0.964	0.972	0.975
1	Hybrid	0.800	0.962	0.968	0.971
0.3	AE-SVM	0.672	0.745	0.785	0.814
0.3	RBF-SVM	0.676	0.784	0.799	0.810
0.3	GP	0.645	0.727	0.790	0.801
0.3	RF	0.753	0.798	0.828	0.834
0.3	Hybrid	0.461	0.771	0.813	0.833

TABLE III
MCC AGAINST ATTACK STRENGTH s AT $n = 1400$, AND GP RECALL PER ATTACK TYPE.

s	0.05	0.1	0.2	0.3	0.5	1.0
AE-SVM	0.47	0.61	0.73	0.81	0.89	0.96
RBF-SVM	0.49	0.62	0.73	0.81	0.90	0.97
GP	0.51	0.62	0.71	0.80	0.90	0.96
RF	0.48	0.62	0.75	0.83	0.92	0.97
GP recall, PNS	0.09	0.18	0.48	0.70	0.88	0.99
GP recall, Blinding	0.85	0.97	1.00	1.00	1.00	1.00
GP recall, Trojan	0.09	0.12	0.25	0.45	0.67	0.84
GP recall, Int.-res.	0.84	0.99	1.00	1.00	1.00	1.00

Table V isolates this bias. When labels depend on direction only, AE-SVM learns from far fewer samples than any baseline. When labels depend on norm only, it stays at chance. QKD telemetry lies between these extremes.

D. Calibration and deferral

The GP is the least calibrated supervised model, and isotonic RF the best (Table I, Fig. 4). The GP's probabilities crowd the decision boundary. At $u < 0.3$ it keeps fewer blocks than isotonic RF or AE-SVM at the same accuracy (Table VI). At $u < 0.1$ it keeps none. On this data a recalibrated point estimate makes the better deferral policy.

TABLE IV
AUC ON A HELD-OUT ATTACK TYPE AT $s = 0.3$ (10 SEEDS). THE LAST COLUMN REMOVES THE MAHALANOBIS FEATURE.

Model	PNS	Blinding	Trojan	Int.-res.	Mean	No Mahal.
AE-SVM	0.950	0.971	0.886	1.000	0.952	0.554
AE-SVM (anchored)	0.954	0.968	0.873	1.000	0.949	0.939
RBF-SVM	0.953	0.881	0.856	0.979	0.917	0.940
GP	0.949	0.973	0.850	0.994	0.941	–
RF	0.963	0.997	0.883	0.998	0.960	–
OC-SVM	0.784	1.000	0.545	1.000	0.832	–

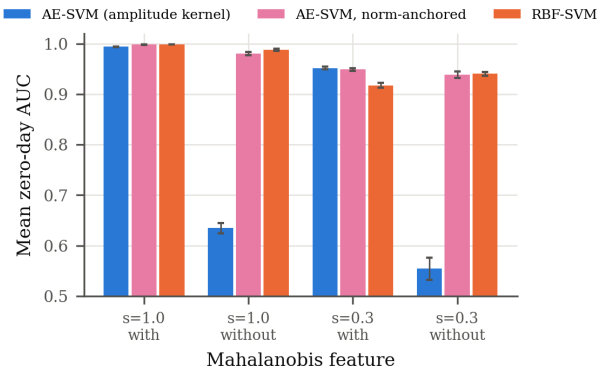


Fig. 3. Mean held-out AUC with and without the Mahalanobis feature.

E. Evasion and cost

A simple evasion lowers RF and GP detection together (Table VII). RF stays 2 to 9 points ahead. The GP's uncertainty barely moves, so it does not flag the evasion. Every detector runs far below a per-block budget measured in seconds (Table VIII).

V. LIMITATIONS

Every result is simulated. The attack sizes are assumptions, and the drift model is not fitted to a real link. The strength axis shows how results move with signature size. It does not say which s a real attacker produces. The evasion is gradient-free and not adaptive. The detectors see single blocks, and models

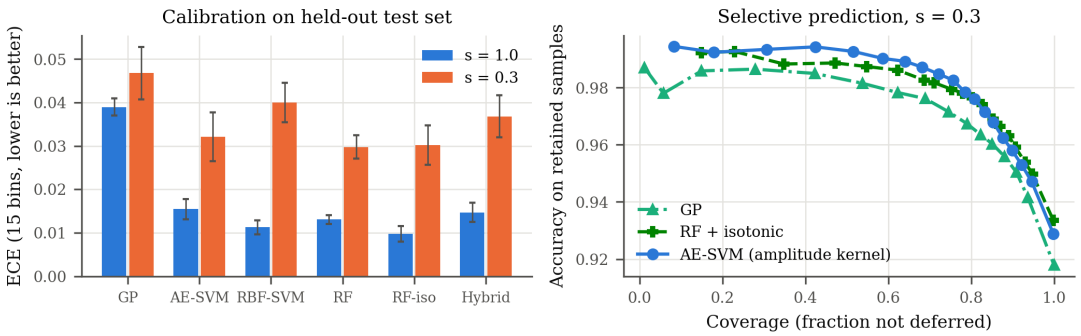


Fig. 4. Left, ECE by model. Right, accuracy against coverage when blocks with uncertainty $u = \sqrt{p(1-p)}$ above a threshold are deferred ($s = 0.3$).

TABLE V
ACCURACY ON CONTROL TASKS IN $\mathbb{R}^8$ WITH HEAVY-TAILED NORMS (10 SEEDS).

Label depends on	n	AE-SVM	RBF-SVM	GP	RF
Direction only	25	0.707	0.556	0.618	0.606
	100	0.856	0.583	0.652	0.667
	1000	0.966	0.744	0.765	0.794
Norm only	25	0.496	0.799	0.691	0.862
	100	0.499	0.855	0.791	0.924
	1000	0.504	0.958	0.950	0.953

TABLE VI
DEFERRAL AT $s = 0.3$. COVERAGE IS THE SHARE OF BLOCKS KEPT, AND ACCURACY IS MEASURED ON THOSE BLOCKS.

	$u < 0.3$		$u < 0.1$	
Model	Coverage	Accuracy	Coverage	Accuracy
GP	69%	97.6%	0%	–
RF-isotonic	80%	97.6%	35%	98.8%
AE-SVM	79%	97.8%	31%	99.3%

over block sequences may do better. Testing on a physical QKD testbed is the next step.

VI. CONCLUSION

On a simulator built not to be trivially separable, random forests are the strongest and fastest-learning detector. Isotonic calibration gives better deferral than a GP posterior. The amplitude kernel is competitive but not better. It generalizes to unseen attacks only when some feature carries magnitude, and a norm anchor supplies this when none does. Simulation studies of QKD detection should report single-feature separability and a strength sweep alongside accuracy.

CODE AND DATA AVAILABILITY

Simulator, models, experiment scripts and all result files are at https://github.com/arnavk23/quasar-security.

TABLE VII
EVASION AT $s = 0.3$. THE FIVE MOST IMPORTANT FEATURES OF EACH ATTACK BLOCK MOVE TOWARD THE NORMAL MEDIAN BY THE GIVEN BUDGET.

Budget	RF detection	GP detection	GP uncertainty
0.0	0.85	0.78	0.30
0.2	0.81	0.73	0.30
0.4	0.74	0.65	0.30
0.6	0.61	0.54	0.30
0.8	0.42	0.40	0.33

TABLE VIII
INFERENCE COST ON ONE CPU CORE, TRAINING EXCLUDED. BATCH COST IS PER BLOCK OVER 600 BLOCKS. SINGLE-BLOCK COST IS DOMINATED BY PYTHON CALL OVERHEAD.

Component	Batch (ms)	Single (ms)
Feature extraction	0.0004	0.14
AE-SVM	0.0084	0.35
RBF-SVM	0.0115	0.16
RF	0.0934	23.35
Hybrid	0.3250	26.85
GP	0.3790	29.63

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